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BATTERY CELL, SECONDARY BATTERY, AND ELECTRIC DEVICE

Abstract

A battery cell includes an electrode assembly, where the electrode assembly includes a first separator, a second separator, and multiple electrode plates stacked. The multiple electrode plates include two outer electrode plates located at two ends and multiple inner electrode plates located between the two outer electrode plates. The first separator is disposed between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate; and the second separator is disposed between adjacent two of the inner electrode plates; where thickness of the first separator is greater than thickness of the second separator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/CN2023/131332, filed on Nov. 13, 2023, which claims priority to Chinese Patent Application No. 202223048292.5, filed on Nov. 16, 2022 and entitled “BATTERY CELL, SECONDARY BATTERY, AND ELECTRIC DEVICE”, each are incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This application relates to the field of battery technologies, and specifically to a battery cell, a secondary battery, and an electric device.

BACKGROUND

[0003] With the development of new energy technologies, the energy storage devices have been increasingly widely used in, for example, mobile phones, laptop computers, battery motorcycles, electric vehicles, electric airplanes, electric ships, electric toy cars, electric toy ships, electric toy airplanes, and electric tools.

[0004] The secondary battery, as an energy storage device, includes a positive electrode plate, a negative electrode plate, an electrolyte, and the like. Lithium ions shuttle back and forth between the positive electrode plate and the negative electrode plate through the electrolyte to achieve charging and discharging. For the secondary battery, both its energy density and safety need to be considered. Therefore, how the safety of the secondary battery is improved is a technical issue to be addressed urgently.

SUMMARY

[0005] Some embodiments of this application provide a battery cell, a secondary battery, and an electric device, to improve the safety performance of the battery cell and secondary battery.

[0006] According to a first aspect, an embodiment of this application provides a battery cell. The battery cell includes an electrode assembly, where the electrode assembly includes a first separator, a second separator, and multiple electrode plates stacked. The multiple electrode plates include two outer electrode plates located at two ends and multiple inner electrode plates located between the two outer electrode plates.

[0007] The first separator is disposed between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate; and the second separator is disposed between adjacent two of the inner electrode plates; where thickness of the first separator is greater than thickness of the second separator.

[0008] In the foregoing technical solution, the first separator is disposed between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate, and the second separator is disposed between adjacent two of the inner electrode plates, meaning that the two first separators are respectively located at the ends of the electrode assembly along the stacking direction of the electrode plates. The thickness of the first separator being greater than the thickness

of the second separator can be understood as follows: compared to the thickness of the separator in the middle region of the electrode assembly, the thickness of the separator at the end of the electrode assembly is increased, which makes the first separator stronger and not prone to puncture, thus reducing the short-circuit risk of the battery cell and improving the safety performance of the battery cell. The larger thickness of the separator (the first separator) at the end of the electrode assembly can also increase the pass rates of the battery cell in the high-voltage test and impact test, reduce the defect rate of the battery cell, and improve the reliability of the battery cell.

[0009] In some embodiments of the first aspect of this application, the thickness of the first separator is $h_{\text{sub.1}}$, and the thickness of the second separator is $h_{\text{sub.2}}$, where $2\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$.

[0010] In the foregoing technical solution, if $h_{\text{sub.1}} - h_{\text{sub.2}} < 2\ \mu\text{m}$, the thickness of the first separator is slightly larger than the thickness of the second separator, the strength of the first separator is not significantly higher than the strength of the second separator, and the first separator is at high risk of being punctured. If $h_{\text{sub.1}} - h_{\text{sub.2}} > 5\ \mu\text{m}$, the thickness of the first separator is significantly larger than the thickness of the second separator, the first separator has greater strength, but the large thickness of the first separator decreases the energy density of the battery cell. Therefore, $2\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$ not only ensures that the first separator has good strength and is not prone to puncture, but also ensures that the increased thickness of the first separator does not significantly decrease the energy density of the battery cell, allowing the energy density of the battery cell to meet the actual requirement.

[0011] In some embodiments of the first aspect of this application, $3\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$.

[0012] In the foregoing technical solution, compared to $2\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 3\ \text{mm}$, in the case of $3\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$, the first separator has a larger thickness and better strength, and is less likely to be punctured, so the short-circuit risk of the battery cell is lower and the battery cell has better safety performance.

[0013] In some embodiments of the first aspect of this application, the thickness of the second separator is $h_{\text{sub.2}}$, where $0\ \mu\text{m} < h_{\text{sub.2}} \leq 7\ \mu\text{m}$.

[0014] In the foregoing technical solution, a thinner separator of the electrode assembly allows the battery cell to have a higher energy density. Therefore, the thickness $h_{\text{sub.2}}$ of the second separator being equal to or smaller than $7\ \mu\text{m}$ is conducive to improving the energy density of the battery cell.

[0015] In some embodiments of the first aspect of this application, $3\ \mu\text{m} \leq h_{\text{sub.2}} \leq 6\ \mu\text{m}$.

[0016] In the foregoing technical solution, if $h_{\text{sub.2}} < 3\ \mu\text{m}$, the thickness of the first separator is small, resulting in weak isolation between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate. If $6\ \mu\text{m} < h_{\text{sub.2}} \leq 7\ \mu\text{m}$, the thickness of the first separator is large, significantly decreasing the energy density of the battery cell. Therefore, $3\ \mu\text{m} \leq h_{\text{sub.2}} \leq 6\ \mu\text{m}$ ensures that the thickness of the first separator meets the requirement for isolation between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate, and also minimizes the impact on the energy density of the battery cell.

[0017] In some embodiments of the first aspect of this application, the thickness of the first separator is $h_{\text{sub.1}}$, and the thickness of the second separator is $h_{\text{sub.2}}$, where $h_{\text{sub.1}} - h_{\text{sub.2}} > 5\ \mu\text{m}$.

[0018] In the foregoing technical solution, compared to $h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$, in the case of $h_{\text{sub.1}} - h_{\text{sub.2}} > 5\ \mu\text{m}$, the thickness increase of the first separator compared to the second separator is larger, and the first separator is stronger and less likely to be punctured, thus further reducing the short-circuit risk of the battery cell and improving the safety performance of the battery cell.

[0019] In some embodiments of the first aspect of this application, the electrode plate includes a current collector, and thickness of the current collectors of some of the multiple electrode plates is smaller than thickness of the current collectors of the other electrode plates.

[0020] In the foregoing technical solution, that thickness of the current collectors of some of the multiple electrode plates is smaller than thickness of the current collectors of the other electrode plates can be understood as follows: the thickness of the current collectors of some of the multiple electrode plates is decreased compared to the thickness of the current collectors of the other electrode plates, which can alleviate the problem of loss in energy density caused by the increased thickness of the first separator compared to the second separator.

[0021] In some embodiments of the first aspect of this application, thickness of the current collector of the outer electrode plate is smaller than thickness of the current collector of the inner electrode plate.

[0022] In the foregoing technical solution, the thickness of the current collector of the outer electrode plate being smaller than the thickness of the current collector of the inner electrode plate is conducive to increasing the pass rate of the battery cell in the high-voltage test.

[0023] In some embodiments of the first aspect of this application, the thickness of the current collector of the outer electrode plate is $m_{sub.1}$, and the thickness of the current collector of the inner electrode plate is $m_{sub.2}$, where $2\ \mu\text{m} \leq m_{sub.2} - m_{sub.1} \leq 5\ \mu\text{m}$.

[0024] In the foregoing technical solution, if $m_{sub.2} - m_{sub.1} < 2\ \mu\text{m}$, a thickness decrease of the current collector of the outer electrode plate compared to the current collector of the inner electrode plate is quite small, resulting in a weak ability of alleviating loss in the energy density caused by the decreased thickness of the first separator compared to the second separator. If $m_{sub.2} - m_{sub.1} > 5\ \mu\text{m}$, the thickness decrease of the current collector of the outer electrode plate compared to the current collector of the inner electrode plate is quite large, leading to a weak strength of the current collector of the outer electrode plate, and leaving the outer electrode plate easy to break and difficult to pass the high-voltage test and impact test. Therefore, $2\ \mu\text{m} \leq m_{sub.2} - m_{sub.1} \leq 5\ \mu\text{m}$ not only ensures that the outer electrode plate has good strength, but also ensures that the thickness decrease of the current collector of the outer electrode plate compared to the current collector of the inner electrode plate can significantly alleviate the problem of loss in the energy density caused by the increased thickness of the first separator compared to the second separator.

[0025] In some embodiments of the first aspect of this application, the second separator includes a connecting section and multiple body sections, the multiple body sections are arranged along a stacking direction of the electrode plates, adjacent two of the inner electrode plates are separated by the body section, and the connecting section connects adjacent two of the body sections.

[0026] In the foregoing technical solution, the second separator includes the connecting section and multiple body sections, and two adjacent body sections are connected by the connecting section, making the second separator a continuous structure. This facilitates the assembly and formation of the electrode assembly and allows the second separator to better separate any two adjacent inner electrode plates.

[0027] According to a second aspect, an embodiment of this application provides a secondary battery including the battery cell provided in any one of the embodiments of the first aspect.

[0028] In the foregoing technical solution, the battery cell provided in any one of the embodiments of the first aspect has low short-circuit risk and good safety and reliability, and the secondary battery including such battery cell also has good safety and reliability.

[0029] According to a third aspect, an embodiment of this application provides an electric device including the secondary battery provided in the embodiment of the second aspect.

[0030] In the foregoing technical solution, the secondary battery provided in the embodiment of the second aspect has good safety and reliability, and can improve the electrical safety of the electric device powered by the secondary battery. The secondary battery can stably and reliably supply power to the electric device.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0031] In order to describe the technical solutions in the embodiments of this application more clearly, the following briefly describes the accompanying drawings required for describing the embodiments. It should be appreciated that, the accompanying drawings below only show some embodiments of this application, and thus should not be considered as a limitation to the scope. A person of ordinary skill in the art may still derive other related drawings from the accompanying drawings without creative efforts.

[0032] FIG. 1 is a cross-sectional view of a battery cell according to some embodiments of this application;

[0033] FIG. 2 is a cross-sectional view of a battery cell according to some other embodiments of this application; and

[0034] FIG. 3 is a cross-sectional view of a battery cell according to still some other embodiments of this application.

[0035] Reference signs: **100**. battery cell; **10**. electrode assembly; **11**. first separator; **12**. second separator; **121**. connecting section; **122**. body section; **13**. electrode plate; **131**. outer electrode plate; **132**. inner electrode plate; **133**. current collector; **134**. active substance layer; **13a**. positive electrode plate; **13b**. negative electrode plate; **14**. third separator; **20**. housing; X. first direction; and Y. second direction.

DETAILED DESCRIPTION

[0036] To make the objectives, technical solutions, and advantages of the embodiments of this application clearer, the following clearly and completely describes the technical solutions in the embodiments of this application with reference to the accompanying drawings in the embodiments of this application. Apparently, the described embodiments are some but not all of the embodiments of this application. Generally, the components of the embodiments of this application as described and illustrated in the accompanying drawings herein can be arranged and designed in a variety of configurations.

[0037] Therefore, the following detailed description of the embodiments of this application as provided in the accompanying drawings is not intended to limit the scope of this application but merely to represent selected embodiments of this application. All other embodiments obtained by persons of ordinary skill in the art based on the embodiments of this application without creative efforts shall fall within the protection scope of this application.

[0038] It should be noted that, without conflict, the embodiments and features in the embodiments in this application may be combined with each other.

[0039] It should be noted that similar reference signs and letters indicate similar items in the following drawings, and therefore once an item is defined in one drawing, it does not need to be further defined and explained in the subsequent drawings.

[0040] In the description of the embodiments of this application, it should be noted that the orientations or positional relationships as indicated are orientations or positional relationships based on the accompanying drawings, or conventional orientations or positional relationships of products of this application, or orientations or positional relationships as conventionally understood by persons skilled in the art, and the orientations or positional relationships as indicated are merely for ease and brevity of description of this application rather than indicating or implying that the apparatuses or elements mentioned must have specific orientations or must be constructed or manipulated according to specific orientations, and therefore cannot be understood as limitations on this application. In addition, the terms “first”, “second”, “third”, and the like are merely intended for distinguishing purposes and shall not be understood as any indication or implication of relative importance.

[0041] Currently, from a perspective of market development, application of secondary batteries is becoming more extensive. The secondary batteries have been widely used in electric transportation vehicles such as electric bicycles, electric motorcycles, and electric cars, and various field such as electric tools, drones, and energy storage devices. With the continuous expansion of application fields of the secondary batteries, market demands for the secondary batteries are also increasing.

[0042] In terms of connection method, the secondary battery may be a single battery cell or a battery pack formed by multiple battery cells connected in series, parallel, or series-parallel. Being connected in series-parallel means that the battery cells are electrically connected in parallel and series. In terms of type, the secondary batteries may include but are not limited to a lithium metal secondary battery, a lithium-ion secondary battery (lithium-ion battery), a sodium-ion secondary battery, a lithium polymer secondary battery, and a lithium-ion polymer secondary battery.

[0043] The battery cell includes an electrode assembly, and the electrode assembly may be a wound electrode assembly or a stacked electrode assembly. The electrode assembly includes a separator and multiple electrode plates stacked, and the separator is disposed between two electrode plates with opposite polarities to separate them.

[0044] The inventors have found that decreasing the thickness of the separator allows the battery cell to have a high energy density, but as the separator gets thinner, the strength of the separator becomes weaker. As a result, the separator is easily punctured, leading to short circuit of the battery cell and low safety performance of the battery cell, especially in the stacking direction of the electrode plates. The decreasing thickness of the separator also exacerbates hardware issues of the battery cell, including low pass rates in the high-voltage test and impact test. This not only increases the defect rate of the battery cell during production, but also reduces the reliability of the battery cell during use. For example, it is difficult for the battery cell to withstand high voltage, and the battery cell has a weak ability of resisting external impact.

[0045] Based on the foregoing considerations, to alleviate the problems of short circuit and increased defect rate of the battery cell caused by the fact that the separator has a small thickness and is easily punctured, an embodiment of this application provides a battery cell, where an electrode assembly of the battery cell includes a first separator, a second separator, and multiple electrode plates stacked. The multiple electrode plates include two outer electrode plates located at two ends and multiple inner electrode plates located between the two outer electrode plates. The first separator is disposed between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate; and the second separator is disposed between adjacent two of the inner electrode plates; where thickness of the first separator is greater than thickness of the second separator.

[0046] The first separator is disposed between the outer electrode plate and the inner electrode plate adjacent to the outer electrode plate, and the second separator is disposed between adjacent two of the inner electrode plates, meaning that the two first separators are respectively located at the ends of the electrode assembly along the stacking direction of the electrode plates. The thickness of the first separator being greater than the thickness of the second separator can be understood as follows: compared to the thickness of the separator in the middle region of the electrode assembly, the thickness of the separator at the end of the electrode assembly is increased, which makes the first separator stronger and not prone to puncture, thus reducing the short-circuit risk of the battery cell and improving the safety performance of the battery cell.

[0047] In addition, the larger thickness of the separator (the first separator) at the end of the electrode assembly can also increase the pass rates of the battery cell in the high-voltage test and impact test, reduce the defect rate of the battery cell, and improve the reliability of the battery cell.

[0048] The battery cell disclosed in this embodiment of this application may be used in, without limitation to, electric devices such as electric two-wheelers, electric tools, drones, and energy storage devices. The battery with the operating condition described in this application may be used as the power system for electric device. This is conducive to improving the safety performance of

the battery cell as well as the electrical safety and reliability of the electric device.

[0049] An embodiment of this application provides an electric device that uses the secondary battery as the power supply. The electric device may be, but is not limited to, an electronic device, an electric tool, an electric vehicle, a drone, or an energy storage device. The electronic device may include a mobile phone, a tablet, a laptop, or the like, the electric tool may include an electric drill, an electric saw, or the like, and the electric vehicle may include an electric car, an electric motorcycle, an electric bicycle, or the like.

[0050] As shown in FIGS. 1 to 3, an embodiment of this application provides a battery cell **100**. The battery cell **100** includes an electrode assembly **10**, and the electrode assembly **10** includes a first separator **11**, a second separator **12**, and multiple electrode plates **13** stacked. The multiple electrode plates **13** include two outer electrode plates **131** located at two ends and multiple inner electrode plates **132** located between the two outer electrode plates **131**. The first separator **11** is disposed between the outer electrode plate **131** and the inner electrode plate **132** adjacent to the outer electrode plate; and the second separator **12** is disposed between adjacent two of the inner electrode plates **132**; where thickness of the first separator **11** is greater than thickness of the second separator **12**.

[0051] The electrode assembly **10** is a laminated electrode assembly **10**. The electrode plate **13** includes a current collector **133** and an active substance layer **134**, and the active substance layer **134** is applied to a surface of the current collector **133** along a thickness direction of the electrode plate **13**. Some of the multiple electrode plates **13** are positive electrode plates **13a**, and the other ones are negative electrode plates **13b**. The positive electrode plate **13a** and the negative electrode plate **13b** are alternately stacked along a first direction X, the first direction X being parallel to the thickness direction of the electrode plate **13**.

[0052] The battery cell **100** mainly relies on movement of metal ions between the positive electrode plate **13a** and the negative electrode plate **13b** to work. The positive electrode plate **13a** includes a positive electrode current collector and a positive electrode active substance layer. The positive electrode active substance layer is applied to a surface of the positive electrode current collector. In some embodiments, the positive electrode current collector may further include a portion uncoated with the positive electrode active substance layer. The portion of the positive electrode current collector uncoated with the positive electrode active substance layer bulges out of the other portion of the positive electrode current collector coated with the positive electrode active substance layer, and the portion of the positive electrode current collector uncoated with the positive electrode active substance layer is used as a positive electrode tab. In some other embodiments, the positive electrode tab may alternatively be a structure that is separated from the positive electrode current collector and then electrically connected to it. Taking a lithium-ion battery as an example, the positive electrode current collector may be made of aluminum, and the positive electrode active substance may be lithium cobaltate, lithium iron phosphate, ternary material, lithium manganate, or the like. The negative electrode plate **13b** includes a negative electrode current collector and a negative electrode active substance layer. The negative electrode active substance layer is applied to a surface of the negative electrode current collector. In some embodiments, the negative electrode current collector further includes a portion uncoated with the negative electrode active substance layer. The portion of the negative electrode current collector uncoated with the negative electrode active substance layer bulges out of the other portion of the negative electrode current collector coated with the negative electrode active substance layer, and the portion of the negative electrode current collector uncoated with the negative electrode active substance layer is used as a negative electrode tab. In some other embodiments, the negative electrode tab may alternatively be a structure that is separated from the negative electrode current collector and then electrically connected to it. The negative electrode current collector may be made of copper, and the negative electrode active substance may be carbon, silicon, or the like. To allow a large current to pass through without any fusing, multiple positive electrode tabs are

provided and stacked together, and multiple negative electrode tabs are provided and stacked together.

[0053] The two outer electrode plates **131** may both be positive electrode plates **13a** or negative electrode plates **13b**. Alternatively, one of the two outer electrode plates **131** is a positive electrode plate **13a** and the other is a negative electrode plate **13b**. The active substance layer **134** may be applied to two surfaces of the current collector **133** of the outer electrode plate **131** along the thickness direction of the electrode plate **13**. Alternatively, the active substance layer **134** may be applied to only a surface facing the inner electrode plate **132**. In FIG. 1, the two outer electrode plates **131** are both positive electrode plates **13a** and the active substance layer **134** is applied to only surfaces of the current collectors **133** of the two outer electrode plates **131** facing the surfaces of the inner electrode plates **132**, so as to avoid waste of the active substance layer **134**.

[0054] The multiple inner electrode plates **132** include positive electrode plates **13a** and negative electrode plates **13b**, and the active substance layer **134** is applied to two opposite surfaces of the current collector **133** of the inner electrode plate **132** along the thickness direction of the electrode plate **13**.

[0055] The first separator **11** and the second separator **12** may be made of a same material. For example, the material of both the first separator **11** and the second separator **12** is PP (polypropylene, polypropylene) or PE (polyethylene, polyethylene). It should be noted that the first separator **11** and the second separator **12** being made of the same material does not mean that proportions of individual components of the material of the first separator **11** and the second separator **12** are exactly the same.

[0056] The first separator **11** and the second separator **12** may alternatively be made of different materials. For example, the first separator **11** is made of PP and the second separator **12** is made of PE, or the first separator **11** is made of PE and the second separator **12** is made of PP. The thickness of the first separator **11** refers to the size of the first separator **11** along the first direction X, and the thickness of the second separator **12** refers to the size of a portion of the second separator **12** between two adjacent inner electrode plates **132** along the first direction X. As shown in FIG. 1, the thickness of the first separator **11** is $h_{sub.1}$, and the thickness of the second separator **12** is $h_{sub.2}$, where $h_{sub.1} < h_{sub.2}$.

[0057] The first separator **11** is disposed between the outer electrode plate **131** and the inner electrode plate **132** adjacent to the outer electrode plate, and the second separator **12** is disposed between adjacent two of the inner electrode plates **132**, meaning that the two first separators **11** are respectively located at the ends of the electrode assembly **10** along the stacking direction of the electrode plates **13**. The thickness of the first separator **11** being greater than the thickness of the second separator **12** can be understood as follows: compared to the thickness of the separator in the middle region of the electrode assembly **10**, the thickness of the separator at the end of the electrode assembly **10** is increased, which makes the first separator **11** stronger and not prone to puncture, thus reducing the short-circuit risk of the battery cell **100** and improving the safety performance of the battery cell **100**. The larger thickness of the separator (the first separator **11**) at the end of the electrode assembly **10** can also increase the pass rates of the battery cell **100** in the high-voltage test and impact test, reduce the defect rate of the battery cell **100**, and improve the reliability of the battery cell **100**.

[0058] It should be noted that the reason why the second separator **12** is punctured in the related art may be burrs on the current collector **133** of the inner electrode plate **132** adjacent to the outer electrode plate **131**, the active substance layer **134** unevenly applied to the inner electrode plate **132** adjacent to the outer electrode plate **131**, and other particulate substance absorbed onto the inner electrode plate **132** adjacent to the outer electrode plate **131** due to static electricity. These factors can puncture the second separator **12** directly or puncture the second separator **12** upon impact to the battery cell **100**.

[0059] The battery cell **100** further includes a housing **20**, and an accommodating space is formed

in the housing **20** for accommodating the electrode assembly **10**. The housing **20** may further accommodate an electrolyte or a solid electrolyte. The material of the housing **20** includes but is not limited to aluminum or steel.

[0060] As shown in FIG. **1**, in some embodiments, the thickness of the first separator **11** is $h_{sub.1}$, and the thickness of the second separator **12** is $h_{sub.2}$, where $2\ \mu\text{m} \leq h_{sub.1} - h_{sub.2} \leq 5\ \mu\text{m}$. In some embodiments, $3\ \mu\text{m} \leq h_{sub.1} - h_{sub.2} \leq 5\ \mu\text{m}$.

[0061] Specifically, $h_{sub.1} - h_{sub.2}$ may be $2\ \mu\text{m}$, $2.5\ \mu\text{m}$, $3\ \mu\text{m}$, $3.5\ \mu\text{m}$, $4\ \mu\text{m}$, $4.5\ \mu\text{m}$, $5\ \mu\text{m}$, or the like. Certainly, $h_{sub.1} - h_{sub.2}$ may alternatively be another value unlisted in the range of $2\ \mu\text{m} - 5\ \mu\text{m}$.

[0062] If $h_{sub.1} - h_{sub.2} < 2\ \mu\text{m}$, the thickness of the first separator **11** is slightly larger than the thickness of the second separator **12**, the strength of the first separator **11** is not significantly higher than the strength of the second separator **12**, and the first separator **11** is at high risk of being punctured. If $h_{sub.1} - h_{sub.2} > 5\ \mu\text{m}$, the thickness of the first separator **11** is significantly larger than the thickness of the second separator **12**, the first separator **11** has greater strength, but the large thickness of the first separator **11** decreases the energy density of the battery cell **100**. Therefore, $2\ \mu\text{m} \leq h_{sub.1} - h_{sub.2} \leq 5\ \mu\text{m}$ not only ensures that the first separator **11** has good strength and is not prone to puncture, but also ensures that the increased thickness of the first separator **11** does not significantly decrease the energy density of the battery cell **100**, allowing the energy density of the battery cell **100** to meet the actual requirement.

[0063] Certainly, in some other embodiments, the thickness of the first separator **11** is $h_{sub.1}$, and the thickness of the second separator **12** is $h_{sub.2}$, where $h_{sub.1} - h_{sub.2} > 5\ \mu\text{m}$.

[0064] Specifically, $h_{sub.1} - h_{sub.2}$ may be $5.5\ \mu\text{m}$, $6\ \mu\text{m}$, $6.5\ \mu\text{m}$, $7\ \mu\text{m}$, $7.5\ \mu\text{m}$, $8\ \mu\text{m}$, or the like. $h_{sub.1} - h_{sub.2}$ may alternatively be another value unlisted within the range greater than $5\ \mu\text{m}$.

[0065] Compared to $h_{sub.1} - h_{sub.2} \leq 5\ \mu\text{m}$, in the case of $h_{sub.1} - h_{sub.2} > 5\ \mu\text{m}$, the thickness increase of the first separator **11** compared to the second separator **12** is larger, and the first separator **11** has better strength and is less likely to be punctured, thus further reducing the short-circuit risk of the battery cell **100** and improving the safety performance of the battery cell **100**.

[0066] In some embodiments, the thickness of the second separator **12** is $h_{sub.2}$, where $0\ \mu\text{m} < h_{sub.2} \leq 7\ \mu\text{m}$. In some embodiments, $3\ \mu\text{m} \leq h_{sub.2} \leq 6\ \mu\text{m}$.

[0067] Specifically, $h_{sub.2}$ may be $1\ \mu\text{m}$, $1.5\ \mu\text{m}$, $2\ \mu\text{m}$, $2.5\ \mu\text{m}$, $3\ \mu\text{m}$, $3.5\ \mu\text{m}$, $4\ \mu\text{m}$, $4.5\ \mu\text{m}$, $5\ \mu\text{m}$, $5.5\ \mu\text{m}$, $6\ \mu\text{m}$, $6.5\ \mu\text{m}$, $7\ \mu\text{m}$, and the like. $h_{sub.2}$ may alternatively be another value unlisted within the range smaller than $7\ \mu\text{m}$.

[0068] A thinner separator of the electrode assembly **10** is more advantageous for the battery cell **100** to have a higher energy density. Therefore, the thickness $h_{sub.2}$ of the second separator **12** being equal to or smaller than $7\ \mu\text{m}$ is conducive to improving the energy density of the battery cell **100**.

[0069] In this embodiment, under a condition that the thickness $h_{sub.2}$ of the second separator **12** is less than or equal to $7\ \mu\text{m}$, each first separator **11** may have an increase of $2\ \mu\text{m}$ to $5\ \mu\text{m}$ in thickness compared to the second separator **12**.

[0070] The battery cell **100** in this embodiment of this application may be prepared using the following method:

[0071] Preparation of the positive electrode plate **13a**: LiCoO_2 (lithium cobalt oxide), Super-P (conductive carbon black), and PVDF (polyvinylidene difluoride, a binder) are mixed well at a mass ratio of 97.6%:1%:1.4%. Then NMP (N-methylpyrrolidone, N-methylpyrrolidone) is added to and mixed well with the resulting mixture, to produce a slurry with certain fluidity. The slurry is separately applied to two surfaces of a $10\ \mu\text{m}$ -thick metal aluminum foil and a single surface of a $20\ \mu\text{m}$ -thick aluminum foil, with a coating weight of $283.9\ \text{mg}/1540.25\ \text{mm}^2$, followed by drying to obtain flexible positive electrode plates. Next, the positive electrode plates are cold-pressed and slitted to obtain a single-side coated positive electrode plate (a positive electrode plate of which a current collector has only one side coated with the active substance layer along the

thickness direction of the current collector) and a double-side coated positive electrode plate (a positive electrode plate of which a current collector has both sides coated with the active substance layer along the thickness direction of the current collector).

[0072] Preparation of the negative electrode plate **13b**: Graphite, Super-P (conductive carbon black), CMC (carboxymethylcellulose sodium), and an SBR (polymerized styrene butadiene rubber,) emulsion are added to deionized water at a mass ratio of 98%:0.5%:1.1%:0.4%, followed by mixing and stirring to uniformity to obtain a slurry with certain fluidity. The slurry is applied to two surfaces of a 10 μm -thick metal copper foil with a coating weight of 151.3 mg/1540.25 mm², followed by drying to produce a flexible negative electrode plate **13b** which is then cold-pressed and slitted to obtain the negative electrode plate **13b**.

[0073] Preparation of the battery cell **100**: The negative electrode plate **13b**, the double-side coated positive electrode plate, and the single-side coated positive electrode plate are punched into sheets with a specified size using a forming mold. These sheets are then stacked with a first separator (with a thickness of 9 μm) and a second separator (with a thickness of 7 μm), and the resulting stack is hot-pressed to form an electrode assembly **10** with a dimension of 31*39*94 (31 mm thick, 39 mm wide, and 94 mm long). The electrode assembly **10** is then subjected to processes such as welding with electrode tabs, top sealing, side sealing, vacuum drying, electrolyte injection, vacuum packaging, formation, aging, and molding, so as to produce the finished battery cell.

[0074] The battery cell **100** prepared using the foregoing method is subjected to the high-voltage test. The specific test method as follows: All positive electrode tabs of the electrode assembly **10** are welded into a whole which is then welded with an adaptor, and all the negative electrode tabs of the electrode assembly **10** are welded into a whole which is then welded with another adaptor. Then, two probes of a high-voltage tester are respectively brought into contact with the positive and negative electrode terminals of the battery cell **100**, a voltage of 30 V to 50 V is applied, and the internal resistance of the battery cell **100** is measured. If the resistance is greater than 1M ohm, it is determined that the battery cell **100** has passed the high-voltage test; otherwise, it is determined that the battery cell **100** has failed the high-voltage test. In other different embodiments, through testing, the thickness of the first separator **11** can be increased by 2 μm to 5 μm compared to the thickness (h.sub.2=7 μm) of the second separator **12**. After the test, the battery cell **100** prepared in the foregoing different embodiments shows an increase of at least 2% in the pass rate of the high-voltage test compared to the battery cell **100** with both the thickness h.sub.1 of the first separator **11** and the thickness h.sub.2 of the second separator **12** as 7 μm .

[0075] For example, under the condition that the thickness h.sub.1 of the first separator **11** and the thickness h.sub.2 of the second separator **12** are both 7 μm , the battery cell **100** shows a pass rate loss of 5% in the high-voltage test. Under a condition that the thickness h.sub.2 of the second separator **12** is 7 μm and the thickness h.sub.1 of the first separator **11** is 9 μm (compared to the first separator **11**, the second separator **12** has an increase of 2 μm in thickness), the pass rate loss of the battery cell **100** decreases from 5% to 3% in the high-voltage test.

[0076] The battery cell **100** prepared using the foregoing method is subjected to the impact test. The specific test method is as follows: Under a condition that the battery cell **100** is fully charged, in a test environment of 20 $\pm$ 5° C., samples are placed on the test platform, and a rod with a diameter of 15.8 mm is positioned at the center of the wide surface of the sample. For cylindrical, pouch, and prismatic samples, the longitudinal axis of each sample is parallel to the flat surface, and perpendicular to the longitudinal axis of the rod. A hammer with a weight of 9.1 $\pm$ 0.1 kg is vertically dropped in free-fall state from a height of 610 $\pm$ 25 mm onto the intersection of the rod and the sample. If the battery cell **100** does not ignite or explode, it passes the impact test. In other different embodiments, the thickness of the first separator **11** can be increased by 2 μm to 5 μm compared to the thickness (h.sub.2=7 μm) of the second separator **12**. After the test, the battery cell **100** prepared in the foregoing different embodiments shows an increase of at least 5% in the pass rate of the impact test compared to the battery cell **100** with both the thickness h.sub.1 of the first

separator **11** and the thickness $h_{sub.2}$ of the second separator **12** as $7\text{ }\mu\text{m}$.

[0077] For example, under the condition that the thickness $h_{sub.1}$ of the first separator **11** and the thickness of the second separator **12** are both $7\text{ }\mu\text{m}$, the battery cell **100** shows a pass rate loss of 5% in the high-voltage test. Under a condition that the thickness $h_{sub.2}$ of the second separator **12** is $7\text{ }\mu\text{m}$ and the thickness $h_{sub.1}$ of the first separator **11** is $9\text{ }\mu\text{m}$ (compared to the first separator **11**, the second separator **12** has an increase of $2\text{ }\mu\text{m}$ in thickness), the pass rate of the battery cell **100** is increased by 5% in the impact test.

[0078] Because the thickness of the first separator **11** is larger than the thickness of the second separator **12**, the first separator **11** occupies a larger space, which leads to a decrease in the energy density of the battery cell **100**. To alleviate the problem that the large thickness of the first separator **11** decreases the energy density of the battery cell **100**, in some embodiments, thicknesses of the current collectors **133** of some or all of the electrode plates **13** can be reduced.

[0079] For example, in some embodiments, the electrode plate **13** includes a current collector **133**, and thickness of the current collectors **133** of some of the multiple electrode plates **13** is smaller than thickness of the current collectors **133** of the other electrode plates **13**.

[0080] That the thickness of the current collectors **133** of some of the multiple electrode plates **13** is smaller than thickness of the current collectors **133** of the other electrode plates **13** can be understood as follows: the thickness of the current collectors **133** of some of the multiple electrode plates **13** is reduced compared to the thickness of the current collectors **133** of the other electrode plates **13**, which can alleviate the problem of loss in energy density due to the increased thickness of the first separator **11** compared to the second separator **12**. As shown in FIG. 1, the thickness of the current collector **133** of the inner electrode plate **132** is defined as $h_{sub.3}$, and the thickness of the current collector **133** of the outer electrode plate **131** is defined as $h_{sub.4}$.

[0081] Thicknesses of the current collectors **133** of the two outer electrode plates **131** may be the same or different. Thicknesses of the current collectors **133** of all the inner electrode plates **132** may be the same or different.

[0082] The current collectors **133** of one or more of the electrode plates **13** may be thinned. The total decreased thickness of the current collectors **133** of the electrode plates **13** may be the same as or different from the total increased thickness of two first separators **11** compared to the second separator **12**.

[0083] In an embodiment in which the current collectors **133** of multiple electrode plates **13** are thinned, for ease of description, the electrode plate **13** with the thickest current collector **133** is defined as the first electrode plate, and the electrode plate with the current collector **133** thinner than that of the first electrode plate is defined as the second electrode plate. The decreased thickness of the current collector **133** of each second electrode plate compared to the current collector **133** of the first electrode plate may be equal to the decreased thickness of one first separator **11** compared to the second separator **12**. Alternatively, the total decreased thickness of the current collectors **133** of multiple second electrode plates compared to the current collector **133** of the first electrode plate may be equal to the total increased thickness of two first separators **11** compared to the second separator **12**.

[0084] In some embodiments, the thickness of the current collector **133** of the inner electrode plate **132** is smaller than thickness of the current collector **133** of the outer electrode plate **131**. It can be understood that the thickness of the current collector **133** of the inner electrode plate **132** is decreased compared to the thickness of the current collector **133** of the outer electrode plate **131**, that is, $h_{sub.3} < h_{sub.4}$, which allows the lighter and thinner electrode plates **13** to be located in the middle region of the electrode assembly **10** and clamped by two outer electrode plates **131**, thus reducing the risk of the inner electrode plates **132** moving between the two outer electrode plates **131**.

[0085] In some other embodiments, the thickness of the current collector **133** of the outer electrode plate **131** is smaller than thickness of the current collector **133** of the inner electrode plate **132**. It

can be understood that the thickness of the current collector **133** of the outer electrode plate **131** is decreased compared to the thickness of the current collector **133** of the inner electrode plate **132**, that is, $h_{sub.4} < h_{sub.3}$.

[0086] The thickness of the current collector **133** of the outer electrode plate **131** has important effect on the pass rate of the battery cell **100** in the high-voltage test. The thickness of the current collector **133** of the outer electrode plate **131** being smaller than the thickness of the current collector **133** of the inner electrode plate **132** is conducive to increasing the pass rate of the battery cell **100** in high-voltage test.

[0087] Further, the thickness of the current collector **133** of the outer electrode plate **131** is $m_{sub.1}$, and the thickness of the current collector **133** of the inner electrode plate **132** is $m_{sub.2}$, where $2\ \mu\text{m} \leq m_{sub.2} - m_{sub.1} \leq 5\ \mu\text{m}$.

[0088] Specifically, $m_{sub.2} - m_{sub.1}$ may be $2\ \mu\text{m}$, $2.5\ \mu\text{m}$, $3\ \mu\text{m}$, $3.5\ \mu\text{m}$, $4\ \mu\text{m}$, $4.5\ \mu\text{m}$, $5\ \mu\text{m}$, or the like.

[0089] If $m_{sub.2} - m_{sub.1} < 2\ \mu\text{m}$, the thickness decrease of the current collector **133** of the outer electrode plate **131** compared to the current collector **133** of the inner electrode plate **132** is quite small, resulting in a weak ability of alleviating loss in the energy density caused by the decreased thickness of the first separator **11** compared to the second separator **12**. If $m_{sub.2} - m_{sub.1} > 5\ \mu\text{m}$, the thickness decrease of the current collector **133** of the outer electrode plate **131** compared to the current collector **133** of the inner electrode plate **132** is quite large, leading to a weak strength of the current collector **133** of the outer electrode plate **131** and leaving the outer electrode plate **131** easy to break and difficult to pass the high-voltage test and impact test. Therefore, $2\ \mu\text{m} \leq m_{sub.2} - m_{sub.1} \leq 5\ \mu\text{m}$ not only ensures that the outer electrode plate **131** has good strength, but also ensures that the thickness decrease of the current collector **133** of the outer electrode plate **131** compared to the current collector **133** of the inner electrode plate **132** can significantly alleviate the problem of loss in the energy density caused by the increased thickness of the first separator **11** compared to the second separator **12**.

[0090] Certainly, in some other embodiments, the thicknesses of the current collectors **133** of all the electrode plates **13** may be the same.

[0091] As shown in FIG. 1, the battery cell **100** may include multiple second separators **12**, where all the second separators **12** are independent of each other, and all the second separators **12** are stacked along the first direction X. The first separator **11** is independent of the second separator **12**.

[0092] As shown in FIG. 2, in some embodiments, only one second separator **12** may be provided. The second separator **12** includes a connecting section **121** and multiple body sections **122**, the multiple body sections **122** are arranged along a stacking direction of the electrode plates **13**, adjacent two of the inner electrode plates **132** are separated by the body section **122**, and the connecting section **121** connects two adjacent body sections **122**.

[0093] In these embodiments, the thickness $h_{sub.2}$ of the second separator **12** refers to the thickness of the body section **122**. The thicknesses of the body section and the connecting section **121** may be the same or different.

[0094] Two adjacent body sections **122** are connected by one connecting section **121**, and an end of one body section **122** is connected to only one connecting section **121**. The connecting section **121** is connected to an end of the body section along a second direction Y, the first direction X being perpendicular to the second direction Y.

[0095] The body section **122** and the connecting section **121** may be disposed separately and then integrated into a whole structure through bonding or the like to form a continuous second separator **12**. The body section **122** and the connecting section **121** may alternatively be integrally formed.

[0096] The second separator **12** includes a connecting section **121** and multiple body sections **122**, two adjacent body sections **122** are connected by the connecting section **121**, making the second separator **12** a continuous structure. This facilitates the assembly and formation of the electrode assembly **10**, and allows the second separator **12** to better separate any two adjacent inner electrode

plates **132**.

[0097] As shown in FIG. 3, in some embodiments, the electrode assembly **10** further includes two third separators **14**, and the two third separators **14** are respectively disposed on sides of the two outer electrode plates **131** back away from the inner electrode plates **132** along the first direction X. The third separator **14** is configured to separate the outer electrode plate **131** from the housing **20** along the first direction X, to reduce the safety risk of electrical leakage caused by the contact between the outer electrode plate **131** and the housing **20**.

[0098] An embodiment of this application further provides a secondary battery, where the second battery includes the battery cell **100** provided in any one of the foregoing embodiments.

[0099] The secondary battery may include multiple battery cells **100**, and the multiple battery cells **100** are connected in series, parallel, or series-parallel via a busbar (not shown in the figure).

[0100] The battery cell **100** provided in any one of the foregoing embodiments has low risk of short circuit as well as good safety and reliability, and the battery including such battery cell **100** also has good safety and reliability.

[0101] An embodiment of this application further provides an electric device, where the electric device includes the secondary battery provided in the foregoing embodiment. The secondary battery serves as a power source for the electric device, enabling the electric device to perform its own function.

[0102] The electric device may be but is not limited to a mobile phone, a tablet, a laptop, an electric drill, an electric saw, an electric car, an electric motorcycle, an electric bicycle, or the like.

[0103] The secondary battery provided in the foregoing embodiment has good safety and reliability, and can improve the electrical safety of the electric device powered by such secondary battery, and the secondary battery can also stably and reliably supply power to the electric device.

[0104] The foregoing descriptions are merely preferred embodiments of this application, but are not intended to limit this application. Persons skilled in the art understand that this application may have various modifications and variations. Any modifications, equivalent replacements, and improvements made without departing from the spirit and principle of this application shall fall within the protection scope of this application.

Claims

1. A battery cell, comprising an electrode assembly, wherein the electrode assembly comprises: a plurality of electrode plates stacked together, the plurality of electrode plates comprising two outer electrode plates located at two ends and a plurality of inner electrode plates located between the two outer electrode plates; a first separator disposed between each outer electrode plate and an inner electrode plate adjacent to the each outer electrode plate; and a second separator disposed between adjacent two of the inner electrode plates; wherein a thickness of the first separator is greater than thickness of the second separator.

2. The battery cell according to claim 1, wherein the thickness of the first separator is $h_{\text{sub.1}}$, and the thickness of the second separator is $h_{\text{sub.2}}$, wherein $2\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$.

3. The battery cell according to claim 2, wherein $3\ \mu\text{m} \leq h_{\text{sub.1}} - h_{\text{sub.2}} \leq 5\ \mu\text{m}$.

4. The battery cell according to claim 1, wherein the thickness of the second separator is $h_{\text{sub.2}}$, wherein $0\ \mu\text{m} \leq h_{\text{sub.2}} \leq 7\ \mu\text{m}$.

5. The battery cell according to claim 4, wherein $3\ \mu\text{m} \leq h_{\text{sub.2}} \leq 6\ \mu\text{m}$.

6. The battery cell according to claim 1, wherein the thickness of the first separator is $h_{\text{sub.1}}$, and the thickness of the second separator is $h_{\text{sub.2}}$, wherein $h_{\text{sub.1}} - h_{\text{sub.2}} > 5\ \mu\text{m}$.

7. The battery cell according to claim 1, wherein, each electrode plate comprises a current collector; and a thickness of the current collectors of some of the plurality of electrode plates is smaller than a thickness of the current collectors of the other of the plurality of electrode plates.

8. The battery cell according to claim 7, wherein, a thickness of the current collectors of the two

outer electrode plates is smaller than a thickness of the current collectors of the plurality of inner electrode plates.

9. The battery cell according to claim 8, wherein the thickness of the current collector of the two outer electrode plates is $m.sub.1$, and the thickness of the current collectors of the plurality of inner electrode plates is $m.sub.2$, wherein $2\ \mu m \leq m.sub.2 - m.sub.1 \leq 5\ \mu m$.

10. The battery cell according to claim 1, wherein the second separator comprises a connecting section and a plurality of body sections; the plurality of body sections are arranged along a stacking direction of the electrode plates; two adjacent inner electrode plates are separated by the body section, and the connecting section connects two adjacent body sections.

11. The battery cell according to claim 1, wherein each electrode plate includes a current collector and an active substance layer, and the active substance layer is applied to a surface of the current collector along a thickness direction of the electrode plate; the active substance layer may be applied to only a surface of the outer electrode plate facing the inner electrode plate.

12. The battery cell according to claim 11, wherein the active substance layer is applied to two opposite surfaces of the current collector of the plurality of inner electrode plates along the thickness direction of the electrode plate.

13. The battery cell according to claim 1, wherein each electrode plate includes a current collector and an active substance layer, a thicknesses of the current collectors of the two outer electrode plates be the same.

14. The battery cell according to claim 1, wherein each electrode plate includes a current collector and an active substance layer, a thicknesses of the current collectors of all the inner electrode plates be the same.

15. The battery cell according to claim 7, wherein the current collectors of more of the electrode plates may be thinned; the total decreased thickness of the current collectors of the electrode plates be the same as the total increased thickness of two first separators compared to the second separator.

16. The battery cell according to claim 7, wherein a thickness of the current collectors of the plurality of inner electrode plate is smaller than a thickness of the current collectors of the two outer electrode plates.

17. The battery cell according to claim 1, wherein the battery cell further includes a housing, and an accommodating space is formed in the housing for accommodating the electrode assembly; the material of the housing includes aluminum or steel.

18. The battery cell according to claim 17, wherein the electrode assembly further includes two third separators, and the two third separators are respectively disposed on sides of the two outer electrode plates back away from the inner electrode plates along the first direction; the third separator is configured to separate the outer electrode plate from the housing along the first direction.

19. A secondary battery, comprising the battery cell according to claim 1.

20. An electric device, comprising the secondary battery according to claim 19.
